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PROFESSIONAL SUMMARY

Information Communication & Technology for Development (ICT4D) professional with 20+ years of experience leading and delivering digital transformation initiatives across government, international NGOs, and donor-funded programs (USAID, WHO, FHI 360, RTI, MSI). Proven track record in building scalable digital systems, managing cross-functional teams, and translating complex field problems into practical technology solutions.

Expertise spans technical leadership, system architecture, data governance, MIS development, and AI-assisted solutions, with strong hands-on capabilities in both engineering and operational environments.

CORE STRENGTHS

- Technical Leadership & System Architecture
- ICT4D & Digital Transformation (Health, Governance, Public Sector)
- Full-stack Web Development & MIS Systems
- Data Systems, Dashboards & Reporting Automation
- QA, Testing, and System Validation (Test IO experience)
- AI Training, Prompt Engineering & Evaluation
- Cross-country stakeholder collaboration

EXPERIENCE SUMMARY (TECH STACK + IMPACT)

1. Technical Leadership (Strategic & System-Level)

FHI 360 — ICT4D Manager (2021–2025)

🔹 **Key Responsibilities**

- Led stakeholder engagement (government, NGOs, donors) to **identify needs, conduct assessments, and translate problems into digital solutions**
- Led national-scale digital systems across health programs
- Managed cross-functional teams (dev, data, operations)
- Designed **end-to-end system architecture** (MIS, dashboards, web/mobile), including data flow, integration, and scalability
- Developed **digital master plans, technical roadmaps, and system blueprints** aligned with program objectives
- Led full product lifecycle: **requirement** → **prototype** → **development** → **deployment** → **iteration**

- Defined **team roles, SOW, and development workflows**, ensuring quality delivery through QA and testing
- Acted as a bridge between **technical teams and program stakeholders**, ensuring solutions are practical and adoptable

◆ **Key Impact**

- Delivered systems used by **multi-level stakeholders (national to field level)**
- Supported **thousands of users** across programs
- Reduced manual reporting
- Improved **real-time data visibility and decision-making**

◆ **Tech Stack**

- ➢ Backend: PHP, Laravel, CodeIgniter, WordPress, Drupal, Node.js, Next.js, Python, Typescript, Javascript.
- ➢ Database: MySQL, PostgreSQL
- ➢ Cloud/Infra: AWS (basic), VPS, Linux
- ➢ Tools: DHIS2, Tableau/Power BI, REST API
- ➢ AI: OpenAI API, prompt engineering

2. **Hands-on Engineering & System Development**

FHI 360 — ICT4D Advisor
2016 – 2021 | Indonesia
Role: System Developer, Technical Reviewer, Solution Designer
Key Responsibilities:

- Designed and improved digital systems (case management, analytics dashboards, web/mobile tools)
- Conducted system review, debugging, and performance optimization
- Translated user and field requirements into technical system workflows
- Acted as technical reviewer for system quality, usability, and scalability
- Collaborated with stakeholders to refine product functionality

MSI — Case Management System (CMS) Consultant
2015 – 2016 | Indonesia
Role: Software Developer, System Implementer
Key Responsibilities:

- Designed and developed case management system for public institutions
- Implemented system features based on user requirements and feedback
- Performed debugging, system refinement, and performance improvements
- Developed user manuals and technical documentation
- Delivered training sessions for system users

Ombudsman RI — System Analyst & Developer
2015 | Indonesia
Role: System Analyst, Integration Developer
Key Responsibilities:

- Evaluated and redesigned system architecture (SP4N, SIMPeL integration)
- Improved system interoperability and data flow
- Delivered technical recommendations and implementation roadmap

- Supported system enhancement for transparency and accessibility

Kemitraan (PRG) — Communication Specialist (Web & Digital Systems)
2014 – 2015 | Indonesia
Role: Web Developer, Content System Manager
Key Responsibilities:
- Managed and developed web-based communication platforms
- Created and maintained digital content systems (website & media)
- Designed content workflows and publication pipelines
- Supported digital campaigns and media integration

USAID IKAT — ICT Specialist
2014 | Indonesia / ASEAN
Role: Web Developer, System Administrator
Key Responsibilities:
- Developed and maintained Knowledge Resource Center platform
- Managed content system and digital knowledge-sharing platform
- Built and maintained web-based communication systems
- Supported system integration across ASEAN partner organizations

MSI — Complaint Handling System (CHS) Consultant
2013 | Indonesia
Role: Consultant, System Developer
Key Responsibilities:
- Designed and improved complaint handling system (e-CHS)
- Conducted system analysis and workflow design
- Developed system roadmap and implementation strategy
- Delivered training and technical assistance

RTI International — ICT Specialist (EMAS Project)
2012 | Indonesia
Role: System Developer, Technical Support
Key Responsibilities:
- Supported development of mobile and web-based health systems
- Assisted in building referral systems and communication platforms
- Conducted system deployment and technical support
- Delivered training for government health staff

🔷 Tech Stack

- **Frontend :** TypeScript, React, Next.js
- **Backend :** TypeScript, Node.js (Express, Fastify, NestJS), Rust (Axum), PHP (Laravel, CodeIgniter), Python, JavaScript
- **Database & Data :** MySQL, PostgreSQL (CloudSQL, AlloyDB), BigQuery, dbt, Trocco
- **Cloud & Infrastructure :** Google Cloud Platform, AWS (basic), VPS, Linux, Docker, Kubernetes, Istio, Cloudflare
- **API & Integration :** REST, GraphQL, gRPC
- **DevOps & Monitoring :** GitHub, GitHub Actions, ArgoCD, Kustomize, Helm, Terraform, Datadog, Mixpanel, Sentry
- **Event & Messaging :** Google Cloud Pub/Sub

- **Authentication & Security :** Auth0
- **CMS & Web Platforms :** WordPress, Drupal
- **Tools & Productivity :** GitHub Copilot, Figma, Storybook
- **Collaboration & Project Management :** Slack, Discord, Jira, Miro, Confluence

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3. **AI Projects — Independent / Freelance**

   a. **Pentestor (AI Security Assistant)**
   An AI assistant for end-to-end penetration testing, covering scanning, vulnerability identification, risk analysis, and remediation planning. Simplifies security audits with faster and more cost-efficient workflows.
   b. **datAIntegrator (AI Data Migration Agent)**
   An AI-powered co-pilot for migrating data from legacy systems to new architectures with automated field mapping, validation, and SQL/API execution. Reduces manual effort while improving accuracy and speed.
   c. **ASPRI Legislator (AI Governance Assistant)**
   An AI assistant for legislators to manage public aspirations, analyze policies, and support communication workflows. Enables more data-driven and responsive decision-making.
   d. **Labour Advocate (AI Legal Assistant)**
   An AI tool that helps workers understand labor rights, draft complaints, and navigate legal issues. Improves access to legal support in a practical and user-friendly way.
   e. **PoligAmI (AI Matchmaking Agent)**
   An AI-based matchmaking assistant aligned with Islamic values, helping users find compatible partners based on preferences and principles. Designed with a respectful and ethical approach.
   f. **Berani Bersama (AI Health Consultant)**
   An AI consultant providing private and science-based guidance on reproductive health, HIV/STD prevention, and related topics. Supports awareness and preventive care.
   g. **HAI Mom (AI Assistant for Mothers)**
   An AI companion for women across life stages, from menstruation and pregnancy to postpartum care and parenting. Provides practical and accessible guidance.
   h. **Review Obat (AI Product Scanner)**
   An AI tool that verifies the safety and legality of medicines and cosmetics by analyzing packaging and regulatory data. Helps users avoid unsafe or unregistered products.
   i. **Check Halal (AI Halal Scanner)**
   An AI scanner that verifies the halal status of food products through packaging analysis. Supports a modern and convenient halal lifestyle.
   j. **AIr Tanah (AI Groundwater Locator)**
   An AI system that identifies optimal water drilling points using geological and resistivity data. Improves efficiency in groundwater exploration.
   k. **Petani MileniAI (AI Farming Assistant)**

An AI assistant that recommends suitable crops based on land conditions and provides step-by-step guidance from planting to harvest. Designed for modern and urban farmers.

l. **Arah Ikan (AI Fish Finder)**
An AI tool that uses satellite and ocean data to recommend optimal fishing locations. Helps small-scale fishers improve catch efficiency.

m. **Meuneuheun (AI Aquaculture Assistant)**
An AI assistant for fish and shrimp farming, offering daily recommendations, condition monitoring, and disease detection. Supports more productive aquaculture operations.

n. **AI Love Sabang (AI Tourism Assistant)**
An AI travel assistant providing cultural insights, culinary recommendations, and local guidance for visitors in Sabang. Enhances the tourism experience.

o. **PandAI Jalan (AI Travel Planner)**
An AI travel planner for Indonesia, offering route planning, cost estimation, and cultural tips. Helps users plan trips more efficiently and intelligently.

p. **AIsisten Umroh Mandiri (AI Umrah Planner)**
An AI assistant for organizing independent Umrah journeys, covering flights, accommodation, visas, transportation, and rituals. Ensures a structured and well-prepared pilgrimage experience.

q. **CekGu (AI Education Assistant)**
An AI tool that helps teachers design critical-thinking exam questions aligned with curriculum standards. Reduces reliance on easily generated AI answers.

r. **DesAInher (AI Design & Cost Estimator)**
An AI system for building design and automated material cost estimation. Enables faster and more efficient planning for construction projects.

🔷 **Tech Stack:**

- **AI & LLM Engineering Stack:** AI-first architecture, LLM integration (OpenAI, Bedrock), LangChain, RAG, vector databases, prompt engineering, agent orchestration, FastAPI/Node.js, Docker, cloud (GCP/AWS)
- **Data Processing & Pipelines:** ETL pipelines, JSON/SQL transformation, data validation, dbt (basic)
- **Memory & State Management:** Session memory, conversation history, long-term context handling
- **DevOps & Deployment:** Docker, Kubernetes (basic), CI/CD (GitHub Actions), cloud deployment (GCP, AWS)
- **Monitoring & Observability:** Logging, error tracking, prompt/output evaluation, performance monitoring
- **Security & Governance:** Input validation, output filtering, role-based access, AI safety considerations

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4. **Consultancy (System Design, MIS, Governance)**

NIPA Korea — ICT Consultant
2022 – 2025 (Project-based) | Remote : Indonesia Based – South Korea Collaboration
Role: Consultant, AI & Digital System Advisor

Key Responsibilities:

- Conducted needs assessment and stakeholder discussions to identify digital transformation opportunities
- Provided technical advisory on AI, robotics, and digital system adoption for local implementation contexts
- Designed solution concepts, system architecture, and implementation approaches aligned with industry use cases (manufacturing/logistics)
- Supported development of proof of concept (PoC), use case scenarios, and technical documentation
- Acted as liaison between technical teams, business stakeholders, and international partners
- Contributed to project planning, roadmap development, and deployment strategy

City Facilitator (Government Program)
2010 – 2011 | Indonesia
Role: Consultant, System & Planning Facilitator
Key Responsibilities:

- Facilitated development of sanitation planning systems
- Supported data collection, analysis, and reporting workflows
- Assisted local government in strategic planning and system implementation

RTI International — LGMS Specialist
2005 – 2009 | Indonesia
Role: Consultant, Software Developer, MIS Specialist
Key Responsibilities:

- Designed and implemented Management Information Systems (MIS)
- Developed citizen gateway and complaint systems (SMS & web-based)
- Supported system architecture and deployment across districts
- Conducted training and capacity building for government staff

5. **Software & Infrastructure Maintenance & Operations**

World Health Organization (WHO) — ICT Officer
2005 | Indonesia
Role: System Operator, Infrastructure Support
Key Responsibilities:

- Managed communication systems (HF, VHF, satellite systems)
- Maintained network operations and system infrastructure
- Supported emergency communication and system reliability
- Diagnosed and resolved technical system issues

Tech Stack & Impact:

- Desktop systems, LAN networking
- Communication Device: VSAT, Satellite Phone, GPS, VPN, VHF, HF)
- Basic programming (C, early web stack)
- Hardware/software troubleshooting

6. **Software Tester/Reviewer/AI Trainer (Freelance / Remote – Parallel Work)**

Micro1: | Global Projects
2005- NOW
Role: QA Tester, Bug Reporter
Key Responsibilities:

- Performed **functional, exploratory, and regression testing** across web and mobile applications
- Identified, documented, and reported bugs with **clear reproduction steps, severity, and evidence**
- Conducted **cross-browser and cross-device testing** to ensure compatibility and usability
- Collaborated with global teams following **structured QA standards and test cycles**
- Validated fixes and ensured **product quality before release**

Tester Work — Software Tester
2004 – NOW (Freelance / Remote)
Role: QA Tester
Key Responsibilities:
• Executed **manual testing scenarios** for web-based applications
• Reported bugs and usability issues with **detailed documentation**
• Assisted in improving **user experience and system reliability**
• Supported testing cycles and **basic validation processes**

Alignerr — AI Trainer / Evaluator
2005 – NOW: (Remote)
Role: AI Trainer, Response Evaluator
Key Responsibilities:

- Evaluated AI-generated responses based on **accuracy, reasoning, and task completion (pass/fail analysis)**
- Analyzed **model decision trajectories and error patterns** to improve output quality
- Reviewed coding-related responses and ensured **logical correctness and alignment with requirements**
- Contributed to **training datasets and evaluation frameworks** for AI models
  • Provided structured feedback to improve **AI performance in real-world scenarios**

Outlier — AI Trainer / Contributor
2005 – NOW (Remote)
Role: AI Trainer, Data Contributor
Key Responsibilities:
• Trained AI models through **data annotation, validation, and structured feedback**
• Worked on tasks involving **coding, reasoning, and language understanding**
• Ensured data quality aligned with **training guidelines and evaluation standards**
• Contributed to improving **model reliability and response consistency**

7. ACADEMIC EXPERIENCE

Teuku Umar University — Lecturer
2003-2004 | Indonesia
Role: Lecturer, Researcher
Key Responsibilities:

- Delivered undergraduate teaching in engineering and applied systems
- Conducted research on coastal and fisheries systems
- Supported academic and field-based data collection

🔷 CROSS-ROLE AGGREGATED IMPACT
🏥 Systems deployed across health, governance, and development sectors
🌍 Experience with USAID, WHO, FHI 360, RTI, MSI
👥 Supported thousands of end users (government staff, health workers, NGOs)
📊 Built and maintained 20+ digital platforms / MIS systems
⚙️ Reduced manual workflows and improved reporting efficiency significantly
🤖 Recently transitioned into AI + automation + system orchestration

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